

PRACTICAL COURSE WORKBOOK

n8n Self-Hosted Automation

Design a reliable workflow that reduces repetitive work

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a mapped automation with tests and recovery steps
SKILLS	n8n, Self-hosted, AI nodes, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use n8n appropriately in a realistic, bounded task.
- Use Self-hosted appropriately in a realistic, bounded task.
- Use AI nodes appropriately in a realistic, bounded task.

WHO THIS IS FOR

People designing reliable no-code and low-code workflows.

BEFORE YOU BEGIN

- Basic spreadsheet and web-app experience

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

n8n Self-Hosted Automation is a structured, practical course covering n8n, Self-hosted, AI nodes. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 n8n: Workflow design	1. Triggers with n8n 2. Actions with Self-hosted 3. Data mapping with AI nodes
02 Self-hosted: Building	1. Connections with n8n 2. Conditions with Self-hosted 3. Transformations with AI nodes
03 AI nodes: Reliability	1. Errors with n8n 2. Testing with Self-hosted 3. Monitoring with AI nodes
04 n8n: Operations	1. Documentation with n8n 2. Security with Self-hosted 3. Maintenance with AI nodes

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

n8n: Workflow design

Apply n8n through triggers, actions, data mapping.

LESSON 01 / 18 MIN

Triggers with n8n

Learning objectives

- Explain triggers in the context of n8n Self-Hosted Automation.
- Apply n8n to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

triggers, n8n, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a triggers result produced with n8n?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 02 / 22 MIN

Actions with Self-hosted

Learning objectives

- Explain actions in the context of n8n Self-Hosted Automation.
- Apply Self-hosted to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

actions, Self-hosted, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a actions result produced with Self-hosted?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 03 / 26 MIN

Data mapping with AI nodes

Learning objectives

- Explain data mapping in the context of n8n Self-Hosted Automation.
- Apply AI nodes to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

data mapping, AI nodes, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a data mapping result produced with AI nodes?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Self-hosted: Building

Apply Self-hosted through connections, conditions, transformations.

LESSON 04 / 30 MIN

Connections with n8n

Learning objectives

- Explain connections in the context of n8n Self-Hosted Automation.
- Apply n8n to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

connections, n8n, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a connections result produced with n8n?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Conditions with Self-hosted

Learning objectives

- Explain conditions in the context of n8n Self-Hosted Automation.
- Apply Self-hosted to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

conditions, Self-hosted, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a conditions result produced with Self-hosted?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Transformations with AI nodes

Learning objectives

- Explain transformations in the context of n8n Self-Hosted Automation.
- Apply AI nodes to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

transformations, AI nodes, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a transformations result produced with AI nodes?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

AI nodes: Reliability

Apply AI nodes through errors, testing, monitoring.

LESSON 07 / 26 MIN

Errors with n8n

Learning objectives

- Explain errors in the context of n8n Self-Hosted Automation.
- Apply n8n to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

errors, n8n, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a errors result produced with n8n?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Testing with Self-hosted

Learning objectives

- Explain testing in the context of n8n Self-Hosted Automation.
- Apply Self-hosted to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

testing, Self-hosted, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a testing result produced with Self-hosted?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Monitoring with AI nodes

Learning objectives

- Explain monitoring in the context of n8n Self-Hosted Automation.
- Apply AI nodes to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

monitoring, AI nodes, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a monitoring result produced with AI nodes?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

n8n: Operations

Apply n8n through documentation, security, maintenance.

LESSON 10 / 22 MIN

Documentation with n8n

Learning objectives

- Explain documentation in the context of n8n Self-Hosted Automation.
- Apply n8n to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

documentation, n8n, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a documentation result produced with n8n?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Security with Self-hosted

Learning objectives

- Explain security in the context of n8n Self-Hosted Automation.
- Apply Self-hosted to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

security, Self-hosted, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a security result produced with Self-hosted?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Maintenance with AI nodes

Learning objectives

- Explain maintenance in the context of n8n Self-Hosted Automation.
- Apply AI nodes to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

maintenance, AI nodes, automation

Retrieval prompt

In n8n Self-Hosted Automation, which evidence best supports a maintenance result produced with AI nodes?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable n8n Self-Hosted Automation project using n8n, Self-hosted, AI nodes.

DELIVERABLE	A working n8n Self-Hosted Automation artefact plus an evidence-based self-review.
TOOLS	A suitable n8n environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using n8n.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER
State the intended user, need, expected outcome and constraints.
2. INPUTS AND ASSUMPTIONS
List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

API Security Top 10

OWASP Foundation - reviewed 2026-08-21

<https://owasp.org/API-Security/>

HTTP Semantics

IETF - reviewed 2026-08-21

<https://www.rfc-editor.org/rfc/rfc9110>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the n8n scope.